

SERAKHS, Turkmenistan

WMO: 389740

Lat: 36.533N

Lon: 61.217E

Elev: 902

StdP: 14.22

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.9	25.2	9.5	9.2	31.8	14.4	11.8	40.5	20.2	59.1	16.8	56.6	3.5	320	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	105.9	68.3	103.4	67.3	101.2	66.7	71.3	98.5	69.9	97.5	68.7	95.9	6.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.2	86.6	81.6	60.1	80.3	80.7	58.3	75.2	80.2	35.7	99.0	34.4	97.6	33.4	96.3	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	14.6	12.1	DB	14.9	110.7	8.1	2.4	9.1	112.4	4.4	113.8	-0.2	115.2	-6.0	116.9	(4)
(5)			WB	13.6	74.3	7.8	2.1	7.9	75.8	3.4	77.0	-1.0	78.2	-6.6	79.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.7	42.9	45.7	55.3	66.1	77.4	85.5	88.4	85.5	76.7	65.6	53.2	45.0		
	DBStd	18.10	10.39	10.28	9.57	9.08	6.54	4.75	4.23	4.59	5.98	8.30	9.35	9.11		
	HDD50	786	265	181	52	4	0	0	0	0	0	4	69	211		
	HDD65	2741	685	541	327	95	5	0	0	0	2	94	369	623		
	CDD50	6524	46	62	217	487	850	1064	1190	1100	800	488	165	55		
	CDD65	3003	1	2	26	128	390	614	725	635	352	113	15	2		
	CDH74	40788	5	21	300	1343	4647	8927	11158	8943	4023	1233	173	15		
	CDH80	24019	0	3	92	575	2456	5496	7243	5560	2059	482	52	1		
(14) Wind	WSAvg	5.0	4.8	5.1	5.2	5.1	5.4	5.9	5.8	5.3	4.4	4.4	4.4	4.6		
(15) Precipitation	PrecAvg	8.0	1.2	1.3	1.8	1.5	0.5	0.0	0.0	0.0	0.0	0.2	0.5	0.9		
	PrecMax	12.3	3.6	3.1	4.0	4.3	1.9	0.4	0.0	0.2	0.7	1.6	2.1	2.4		
	PrecMin	5.0	0.2	0.2	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.8	0.8	0.8	0.9	1.0	0.6	0.1	0.0	0.0	0.1	0.3	0.5	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.8	77.0	88.3	97.2	104.0	108.1	109.5	107.7	102.0	95.7	86.9	76.5		
		MCWB	52.2	56.1	60.4	64.2	66.7	68.9	69.6	68.4	65.5	63.0	59.1	53.8		
	2%	DB	68.4	71.0	81.7	91.4	99.3	104.4	106.4	104.0	97.9	90.0	78.5	69.5		
		MCWB	50.1	52.1	58.0	63.7	65.9	67.5	69.0	66.9	64.2	61.8	55.8	50.5		
	5%	DB	63.1	66.0	76.2	86.9	95.6	101.7	103.8	101.4	94.4	85.4	73.2	64.5		
		MCWB	48.0	50.4	56.5	61.9	65.0	66.5	67.9	66.4	63.4	59.8	54.3	48.6		
	10%	DB	57.8	61.2	71.1	81.8	92.1	99.2	101.5	98.9	91.0	80.9	68.3	59.6		
		MCWB	46.0	49.0	55.1	60.3	64.1	66.0	67.5	65.5	62.5	58.5	52.5	47.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	57.1	63.0	68.4	71.0	72.6	74.1	72.2	68.5	66.2	60.8	55.4		
		MCDB	69.4	72.3	80.3	86.2	95.4	100.2	100.3	98.6	93.4	89.4	79.3	72.8		
	2%	WB	50.9	54.0	59.7	65.4	68.2	70.0	71.8	70.1	66.5	63.6	58.0	52.1		
		MCDB	64.9	67.5	75.8	84.8	91.6	99.0	99.0	97.0	91.4	85.2	73.6	66.5		
	5%	WB	48.7	51.7	57.8	63.4	66.8	68.7	70.3	68.5	65.0	61.6	55.9	49.8		
		MCDB	61.2	63.0	72.7	81.8	89.3	96.8	98.3	95.9	90.1	81.5	69.8	62.0		
	10%	WB	46.5	49.5	55.8	61.6	65.5	67.5	69.1	67.0	63.5	59.7	53.7	47.7		
		MCDB	56.7	59.7	69.0	78.7	87.0	94.6	97.3	94.9	88.1	77.8	66.4	58.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	17.5	19.3	21.2	23.7	24.3	24.3	25.1	25.3	24.0	19.8	16.5		
		MCDBR	24.4	25.8	28.5	29.2	29.0	28.3	27.1	27.6	29.2	29.9	27.3	24.0		
		MCWBR	11.4	12.1	11.4	10.6	9.7	9.5	9.0	9.8	11.4	13.2	12.3	11.3		
		MCDBR	22.5	22.8	24.3	25.6	25.2	24.6	23.4	23.7	25.5	25.8	23.2	22.0		
	5% WB	MCWBR	11.4	11.8	11.2	10.2	9.1	9.4	9.0	9.6	10.8	12.1	11.6	11.4		
(40) Clear-Sky Solar Irradiance	taub		0.388	0.437	0.508	0.541	0.549	0.547	0.534	0.489	0.458	0.478	0.443	0.392		
	taud		2.094	1.972	1.824	1.794	1.795	1.788	1.821	1.906	1.952	1.892	1.965	2.083		
	Ebn at Noon		244	245	239	239	240	240	242	250	251	228	220	231		
	Edh at Noon		38	49	62	68	69	69	67	60	54	52	44	37		
(44) All-Sky Solar Radiation	RadAvg		731	1006	1365	1796	2267	2578	2533	2353	1956	1382	898	685		
	RadStd		65	121	116	177	83	70	44	56	46	74	83	71		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	+0.77	-0.70	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page